Default Arguments:

A default argument is a default value provided for a function parameter. If the user does not supply an explicit argument for a parameter with a default argument, the default value will be used. If the user does supply an argument for the parameter, the user-supplied argument is used.

Because the user can choose whether to supply a specific argument value, or use the default, a parameter with a default value provided is often called an optional parameter.

Consider the following program:

```
#include<iostream>

void printValues(int ,int =10); // 10 is default argument

int main()

{

        printValues(5); // y will use default argument 10

        printValues(20,35);  //  y will use user-supplied value 35

}


void printValues(int x,int y)

{

        cout<<"x :"<<x<<endl;

        cout<<"y :"<<y<<endl;

}
```

output :

```
        x : 5

        y : 10

        x : 20

        y : 35
```

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In the first function call, the caller did not supply an argument for y, so the function used the default value of 10. In the second call, the caller did supply a value for y, so the user-supplied value was used.

Default arguments are an excellent option when the function needs a value that the user may or may not want to override.

Function supports Multiple default arguments.

Example:

```
#include<iostream>
using namespace std;
void printValues(int =10,int=20,int=30)
int main()
{
printValues(1,2,3);
printValues(1,2);
printValues(1);
printValues( );
}
void printValues(int x,int y,int z)
{
    return x+y+z;
}
```

Key Points:

- Default arguments are different from constant arguments as constant arguments can't be changed whereas default arguments can be overwritten if required.
- Default arguments are overwritten when calling function provides values for them. For example, calling of function sum(10, 15, 25, 30) overwrites the value of z and w to 25 and 30 respectively.

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•During calling of function, arguments from calling function to called function are copied from left to right. Therefore, sum(10, 15, 25) will assign 10, 15 and 25 to x, y, and z. Therefore, the default value is used for w only.

•Once default value is used for an argument in function definition, all subsequent arguments to it must have default value. It can also be stated as default arguments are assigned from right to left. For example, the following function definition is invalid as subsequent argument of default variable z is not default.

•When Function overloading is done along with default values. Then we need to make sure it will not be ambiguous.

The compiler will throw error if ambiguous.

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